

Static Solutions of Einstein's Field Equations for Spheres of Fluid and Application to massive neutron cores

Based on: R. C. Tolman, *Phys. Rev.* **55**, 364–373 (1939) and

Based on: J. R. Oppenheimer & G. M. Volkoff,
Phys. Rev. **55**, 374–381 (1939)

Summary prepared by Gytis Vėjelis

- Solve Einstein's field equations for **static, spherically symmetric** distributions of perfect fluid.
- There are only a few known solutions:
 - Einstein static universe (constant ρ , p everywhere).
 - Schwarzschild interior solution (incompressible fluid with constant ρ).
 - Schwarzschild exterior solution (vacuum outside spherical body).
 - de Sitter solution (vacuum with cosmological constant).

Einstein Equations and Conventions

- Units: $c = G = 1$.
- Einstein equations with cosmological constant Λ :

$$8\pi T_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu}.$$

- Matter: **perfect fluid**

$$T^{\mu\nu} = (\rho + p)u^{\mu}u^{\nu} - pg^{\mu\nu}.$$

- We look for **static, spherically symmetric** configurations in **gravitational equilibrium**.

From the Metric to Einstein Equations

Goal: Given a metric $g_{\mu\nu}$, compute $G_{\mu\nu}$ and solve

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}.$$

1. Start with Static, spherically symmetric ansatz:

$$ds^2 = e^{\nu(r)} dt^2 - e^{\lambda(r)} dr^2 - r^2 d\Omega^2.$$

2. Christoffel symbols

$$\Gamma_{\mu\nu}^{\rho} = \frac{1}{2} g^{\rho\sigma} (\partial_{\mu} g_{\sigma\nu} + \partial_{\nu} g_{\sigma\mu} - \partial_{\sigma} g_{\mu\nu}).$$

3. Riemann tensor

$$R^{\rho}_{\sigma\mu\nu} = \partial_{\mu} \Gamma_{\nu\sigma}^{\rho} - \partial_{\nu} \Gamma_{\mu\sigma}^{\rho} + \Gamma_{\mu\lambda}^{\rho} \Gamma_{\nu\sigma}^{\lambda} - \Gamma_{\nu\lambda}^{\rho} \Gamma_{\mu\sigma}^{\lambda}.$$

4. Ricci tensor and scalar

$$R_{\mu\nu} = R^{\rho}_{\mu\rho\nu}, \quad R = g^{\mu\nu} R_{\mu\nu}.$$

5. Einstein tensor

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}.$$

Perfect Fluid in the Static Frame

- For the RHS we can consider fluids rest frame and obtain

$$T_1^1 = T_2^2 = T_3^3 = -p, \quad T_4^4 = \rho.$$

- Effectively:

$$T^\mu_\nu = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & -p & 0 & 0 \\ 0 & 0 & -p & 0 \\ 0 & 0 & 0 & -p \end{pmatrix}.$$

Reduced Field Equations

Using Tolman's notation the field equations reduce to

$$8\pi\rho = e^{-\lambda} \left(\frac{\lambda'}{r} - \frac{1}{r^2} \right) + \frac{1}{r^2} + \Lambda,$$

$$8\pi p = e^{-\lambda} \left(\frac{\nu'}{r} + \frac{1}{r^2} \right) - \frac{1}{r^2} + \Lambda,$$

$$8\pi p = e^{-\lambda} \left(\frac{\nu''}{2} - \frac{\lambda'\nu'}{4} + \frac{\nu'^2}{4} + \frac{\nu' - \lambda'}{2r} \right) + \Lambda.$$

- Two equations for p must be equal $\Rightarrow$ one relation between λ and ν .
- Together: three ODEs for four unknowns $(\lambda(r), \nu(r), \rho(r), p(r))$.

- Mathematically, we need **one more relation** to close the system. One approach would be to consider some physical choice: **equation of state** $p = p(\rho)$. But this typically leads to non-elementary ODEs needing numerical integration.
- Tolman's idea: Impose a relation between the metric functions $\lambda(r)$ and $\nu(r)$ that makes the system integrable. Consider the physics later

Rewriting the main equation

- Tolman rewrites the equations into **almost** integrable form:

$$\frac{d}{dr} \left(\frac{e^{-\lambda} - 1}{r^2} \right) + \frac{d}{dr} \left(\frac{e^{-\lambda} \nu'}{2r} \right) + e^{-\lambda-\nu} \frac{d}{dr} \left(\frac{e^{\nu} \nu'}{2r} \right) = 0$$

$$8\pi p = e^{-\lambda} \left(\frac{\nu'}{r} + \frac{1}{r^2} \right) - \frac{1}{r^2} + \Lambda$$

$$8\pi \rho = e^{-\lambda} \left(\frac{\lambda'}{r} - \frac{1}{r^2} \right) + \frac{1}{r^2} - \Lambda$$

- Then get solutions by:
 - 1 Choose an **ansatz** relating λ and ν .
 - 2 Integrate the resulting equation for the metric.
 - 3 Use remaining Einstein equations to find $\rho(r)$ and $p(r)$.

Solutions I–VIII

Tolman lists eight families of solutions, labeled I–VIII:

Solution I Einstein static universe: $e^\nu = \text{const}$, spatial curvature radius R .

Solution II Schwarzschild–de Sitter: constant e^λ , includes mass m and Λ .

Solution III Schwarzschild interior: incompressible fluid, constant density ρ .

Solution IV New compressible-fluid model:

$$e^\lambda = \frac{1 + 2r^2/A^2}{1 - r^2/R^2}, \quad e^\nu = B^2(1 + r^2/A^2)$$

with $\rho(r)$ and $p(r)$ finite at $r = 0$.

Solution V Family with $e^\nu \propto r^{2n}$, $e^{-\lambda} = 1 - (1 + 2n - n^2)(r/R)^2$.

Solution VI Another family with $e^{-\lambda} = \text{const}$ and power-law r -dependence in e^ν .

Solution VII More complicated polynomial/reciprocal form (less convenient physically).

Solution VIII More general power-law ansatz, including II as a special case.

Birkhoff's Theorem

Birkhoff's Theorem

Any spherically symmetric solution of the vacuum Einstein equations must be *static* and is uniquely given by the Schwarzschild–(de Sitter) metric.

- The external gravitational field of any spherically symmetric object (star, planet, collapsing matter) is *static* regardless of time-dependent interior dynamics.
- The vacuum exterior is always described by:

$$ds^2 = \left(1 - \frac{2M}{r} - \frac{\Lambda r^2}{3}\right) dt^2 - \frac{dr^2}{1 - \frac{2M}{r} - \frac{\Lambda r^2}{3}} - r^2 d\Omega^2.$$

- Mass parameter M fully characterizes the exterior geometry.

Physical Fluid Spheres and Matching

- Physical model: finite fluid sphere ($0 \leq r \leq r_b$) surrounded by vacuum ($r > r_b$).
- Inside: one of Tolman's interior solutions with $p(r_b) = 0$, $\rho(r_b) > 0$.
- Outside: Schwarzschild metric (neglect Λ on stellar scales)

$$ds^2 = - \left(1 - \frac{2m}{r}\right)^{-1} dr^2 - r^2 d\Omega^2 + \left(1 - \frac{2m}{r}\right) dt^2,$$

where m is the gravitational mass seen at infinity.

- At $r = r_b$, require continuity of the metric and vanishing pressure:

$$p(r_b) = 0, \quad e^{\nu(r_b)} = 1 - \frac{2m}{r_b}, \quad e^{-\lambda(r_b)} = 1 - \frac{2m}{r_b}.$$

Solution IV: Metric, Density, Pressure, Equation of State

- Ansatz: $e^{\nu/2} r^{-1} = \text{const} \Rightarrow e^{\nu/2} \propto r$ (Tolman's relation).
- Resulting interior metric:

$$ds^2 = -\frac{1 + 2r^2/A^2}{1 - r^2/R^2} dr^2 - r^2 d\theta^2 - r^2 \sin^2 \theta d\phi^2 + B^2 \left(1 + \frac{r^2}{A^2}\right) dt^2.$$

- Energy density and pressure (for $\Lambda = 0$):

$$8\pi\rho(r) = \frac{1}{A^2} + \frac{3}{R^2} - \frac{1}{A^2} \frac{1 + 3r^2/R^2}{1 + 2r^2/A^2}, \quad 8\pi p(r) = \frac{1}{A^2} - \frac{1}{R^2} - \frac{1}{A^2} \frac{1 - r^2/R^2}{1 + 2r^2/A^2}.$$

- Central values ($r = 0$):

$$8\pi\rho_c = \frac{3}{R^2} + \frac{1}{A^2}, \quad 8\pi p_c = \frac{1}{A^2} - \frac{1}{R^2}.$$

- Equation of state can be written in a simple quadratic form:

$$(\rho - \rho_c)^2 = \rho_c^2 + 6(\rho_c - p_c)(p_c - p) + 9(p_c - p)^2.$$

(Here A and B are integration constants)

Solution IV: Physical Features

- Parameters A, R control central density/pressure and size.
- For $R^2 > A^2$, pressure and density decrease monotonically with r .
- Pressure vanishes at $r = r_b$:

$$r_b = R \sqrt{1 - \frac{A^2}{R^2}},$$

giving a finite radius.

- Density at boundary remains positive.
- Ratio p/ρ never exceeds $1/3$ if $R^2 \gtrsim 11.5 A^2$ (Tolman's estimate).
- As $A, R \rightarrow \infty$, star radius grows and central density/pressure go to zero.

Solution V: $n = -\frac{1}{2}$ Case

- Take the family with

$$e^{-\lambda} = 1 - (1 + 2n - n^2) \left(\frac{r}{R} \right)^2, \quad e^{\nu/2} \propto r^n.$$

- Special choice $n = -\frac{1}{2}$:

- Central pressure and density $\rightarrow \infty$.
- Ratio at the center:

$$\frac{p_c}{\rho_c} = \frac{1}{3},$$

characteristic of radiation or ultrarelativistic matter.

- Pressure decreases to zero at a finite radius r_b , while density stays finite there.
- Useful toy model for extremely compact objects with radiation-dominated core.

Solution VI: $n = -\frac{1}{2}$ and Blackbody Limit

- Another family with $e^{-\lambda} = \text{const}$ and power-law behavior in e^ν .
- Again, a special choice $n = -\frac{1}{2}$ gives $p_c/\rho_c = 1/3$.
- Equation of state at high density:

$$p - \frac{1}{3}\rho \approx \text{const} \times \rho^{2/3},$$

resembling a highly degenerate Fermi gas at large ρ .

- As certain parameter ratios $\rightarrow 0$, the solution approaches a limiting *blackbody radiation solution* with $p = \rho/3$ throughout.
- Tolman notes close relation to Oppenheimer-Volkoff neutron star models.

Motivation: Beyond Tolman's Analytic Interiors

- Tolman classified analytic interior solutions, but realistic stars obey a *physical equation of state*.
- After nuclear fuel is exhausted, stellar cores may become extremely dense.
- At high densities electrons combine with protons $\Rightarrow$ degenerate **neutron matter**.
- Oppenheimer & Volkoff (1939) study:
 - General relativistic hydrostatic equilibrium (TOV equation).
 - Equation of state of a cold, degenerate Fermi gas of neutrons.
 - Existence of a maximum mass for static neutron cores.

General Relativistic Hydrostatic Equilibrium

Starting from the spherically symmetric metric and perfect fluid:

$$ds^2 = e^{\nu(r)} dt^2 - e^{\lambda(r)} dr^2 - r^2 d\Omega^2,$$

and energy–momentum conservation, Oppenheimer & Volkoff derive the **TOV equation**:

$$\frac{dp}{dr} = - \frac{(\rho + p) [m(r) + 4\pi r^3 p]}{r [r - 2m(r)]},$$

- This is the relativistic generalization of hydrostatic equilibrium.
- Much stronger gravitational effects arise when $r \rightarrow 2m(r)$.

Equation of State: Cold Degenerate Neutron Gas

Oppenheimer & Volkoff use the exact Fermi-gas equation of state (parametrized by t):

$$\rho = K(\sinh t - t), \quad p = \frac{K}{3} \left(\sinh t - 8 \sinh \frac{t}{2} + 3t \right),$$

with

$$K = \frac{m_n^4 c^5}{4\pi^2 \hbar^3}. \quad \text{and} \quad t = 4 \log \left(\frac{\hat{p}}{\mu_0 c} + \left[1 + \frac{\hat{p}}{\mu_0 c} \right] \right)$$

- At low density ($t \ll 1$): Newtonian polytrope $p \propto \rho^{5/3}$.
- At extremely high density ($t \rightarrow \infty$):

$$p \rightarrow \frac{1}{3} \rho c^2,$$

the **ultrarelativistic limit**.

- The EOS becomes stiff but not stiff enough to support arbitrarily large masses.

Solving the Structure Equations

- Choose a central value t_0 (equivalently ρ_c).
- Integrate numerically:

$$\frac{dm}{dr} = r^2(\sinh t - t)$$

$$\frac{dt}{dr} = -\frac{4}{r(r-2m)} \frac{\sinh t - 2 \sinh \frac{1}{2}t}{\cosh t - 4 \cosh \frac{1}{2}t + 3} \times \left[\frac{1}{2}r^3(\sinh t + 8 \sinh \frac{1}{2}t + 3t) \right]$$

until $p(r_b) = 0$.

- This determines the stellar radius r_b and the total gravitational mass $M = m(r_b)$.
- Repeat for many t_0 to obtain $M(t_0)$.

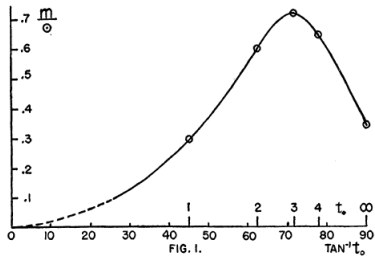


FIG. 1. Dependence of m on t_0 for neutrons.

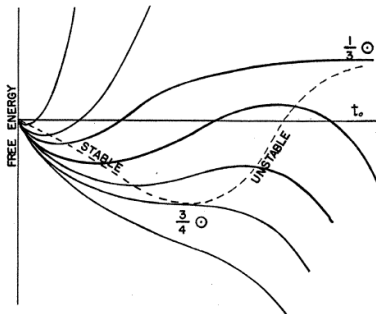


FIG. 3. Schematic plot of free energy as a function of t_0 .

The Oppenheimer-Volkoff Limit

- Numerical integration shows:

$$M_{\text{max}} \approx 0.7 M_{\odot}$$

for a cold neutron gas (Fermi EOS).

- For $M > M_{\text{max}}$:
 - No static solutions.
 - Pressure cannot balance gravity.
 - The core must collapse indefinitely (GR effect).
- The limiting configuration is **not** singular in $g_{\mu\nu}$: GR effects dominate before an event horizon forms.

Physical Interpretation

- At low density: pressure dominated by nonrelativistic neutron degeneracy.
- At high density: neutrons become ultrarelativistic, $p \simeq \rho/3$.
- But relativity strengthens gravity so pressure gravitates
- Pressure grows more slowly than gravity at high density.
- No equation of state of the form $p \leq \rho/3$ can support arbitrarily large mass.

Relation to Tolman's Solutions

Oppenheimer & Volkoff explicitly compare their results with Tolman's analytic classes to understand limiting behavior.

- **Tolman IV:** finite central pressure, realistic density profiles.
 - Exhibits a maximum mass for given central values.
 - Qualitatively reproduces the OV limit.
- **Tolman V and VI:** singular or ultrarelativistic solutions.
 - Approach $p = \rho/3$ at high density.
 - Mirror the behavior near the OV maximum where core becomes radiation-like.

These analytic families help interpret why neutron stars have a maximum mass.

Astrophysical Implications

- Ordinary stars, after exhausting nuclear fuel, may form neutron cores.
- But:

$$M_{\text{core}} > 0.7M_{\odot} \Rightarrow \text{no static equilibrium.}$$

- Therefore massive stars cannot end as static neutron cores.
- They must undergo continued collapse.
- Now we know that this is the formation of black holes

Thank you